

Supporting Dementia Care Through Synthetic Routine Generation and Wearable Anomaly Detection

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Abstract: Dementia often leads to a gradual loss of independence and disruption of daily routines, placing a heavy burden on both patients and caregivers. Wearable devices offer a promising way to monitor activities continuously and non-invasively. However, developing Artificial Intelligence (AI) systems that can reliably detect unusual behaviours remains challenging due to limited real-world data. To address this gap, we propose a methodology for generating synthetic daily routines and introducing anomalies that reflect real situations such as missed meals or nighttime wandering. Using this dataset, we trained an anomaly detection model based on Gated Recurrent Unit (GRUs), which achieved strong results with an F1-score of 0.959 and AUROC of 0.999. These findings demonstrate the potential of synthetic data to overcome current limitations and support the creation of AI-driven monitoring tools for more personalized and proactive dementia care.

1 INTRODUCTION

Dementia affects over 10 million new individuals each year worldwide, equating to a new case every 3.2 seconds (Alzheimer Association, 2025). One of the most critical challenges in dementia care is the progressive loss of autonomy and disruption of daily routines, which significantly impact patient well-being and increase caregiver burden.

Continuous, non-invasive monitoring of physical and behavioural parameters through wearable devices offers a promising approach to addressing these challenges. By capturing real-time data on mobility, sleep, and daily activities, wearables enable early identification of health deterioration, support timely interventions, and improve long-term care outcomes (Freitas et al., 2025). In addition, they can foster greater patient autonomy and alleviate caregiver stress by seamlessly integrating activity recognition and behavioural monitoring into everyday life (Carvalho et al., 2025; Rocha et al., 2024). Their portability, unobtrusiveness, and capacity for longitudinal data collection make them particularly well-suited for dementia care, where continuous routine assessment is essential.

Nevertheless, developing AI-based systems capable of detecting deviations from normal behaviour remains an open challenge. While current activity recognition models are effective for classifying well-defined activities, identifying subtle anomalies such as skipped meals, increased sedentary behaviour, or disrupted routines requires models that can capture long-term temporal dynamics and personalized baselines (Chalapathy & Chawla, 2019). In addition, acquiring sufficiently large and diverse labeled datasets for anomaly detection is particularly difficult in dementia care contexts, due to privacy concerns, ethical constraints, and the rarity and unpredictability of anomalous events (Smuck et al., 2021).

Synthetic data generation provides a viable strategy to overcome these limitations. By simulating realistic daily routines and embedding controlled anomalies, synthetic datasets facilitate the training and evaluation of anomaly detection models without relying solely on extensive real-world data (Liu et al., 2025).

This study builds on prior work in wearable-based monitoring to explore synthetic routine generation combined with AI-driven anomaly detection for personalized dementia care. The proposed system integrates an in-ear sensor, a wrist-worn tracker, environmental beacons, and a base station to continuously

acquire physiological and behavioural signals, including heart rate, body temperature, Global Positioning System (GPS) location, and activity patterns (Arantes et al., 2025). These data streams are processed by an AI engine and visualized through a web portal to support caregivers in home and institutional contexts. In doing so, this work contributes to advancing intelligent, wearable-based systems for monitoring daily routines and detecting behavioural changes in dementia care.

This work is organized in 4 sections. Section 2, Methodology presents the approach for generating synthetic daily routines, simulating anomalies, and implementing the GRU-based detection model. Section 3, Results, reports the performance of the proposed model and includes an analysis of classification metrics. And lastly, Section 4, Conclusion and Future Work, summarizes the main contributions of the study and outlines directions for future research.

2 METHODOLOGY

Synthetic daily routines were generated to provide a structured yet flexible representation of activities across a 24-hour cycle for elderly with dementia. Each day was divided into 288 five-minute intervals, matching the expected operating frequency of the wearable device in development, and segmented into five broad time blocks. Those being morning (7h to 10h), late morning (10h to 12h), afternoon (12h to 18h), evening (18h to 22h), and night (22h to 07h). Within each block, probabilistic rules assigned activities such as meals, hygiene, rest, mobility, and sedentary behaviour, informed by patterns described in dementia care literature.

Each daily activity was associated with a typical duration range, for example, 10 to 30 minutes for hygiene, 15 to 45 minutes for meals, and random variability was introduced to reflect the flexibility of the routine between individuals. This included small shifts in timing, changes in activity order, and variation in duration, allowing the generation of unique but behaviourally plausible daily schedules.

To evaluate the ability of AI models to detect behavioural deviations in daily routines, synthetic anomalies were systematically injected into the generated dataset. The process builds upon the routine generation strategy described earlier and introduces a controlled fraction of anomalous events per day. For example, omitting meals in the afternoon period. These deviations reflect atypical behaviours such as nighttime wandering, missed meals, or activity fragmentation, behaviours often observed in individuals with cognitive impairment.

To evaluate the effectiveness of anomaly detection techniques on the generated dataset, we implemented a sequence model based on Gated Recurrent Units (GRUs) capable of learning temporal dependencies and identifying behavioural deviations.

3 RESULTS

The GRU-based anomaly detection model was evaluated on a synthetic dataset of 20,000 daily sequences of 288 five-minute intervals. Anomalies were injected into 10,000 sequences, affecting 15% of steps per day. Data were split into 70% training, 15% validation, and 15% testing.

The model achieved an Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.999, and an Area Under the Precision–Recall Curve (PR-AUC) of 0.989, and overall accuracy of 99.4%. For normal steps, precision and recall were 0.995 and 0.998. For anomalous steps, the model reached a precision of 0.978 and recall of 0.941, resulting in an F1-score of 0.959. While accuracy shows that the model classifies most steps correctly, AUROC and PR-AUC give a clearer picture when classes are imbalanced. The near-perfect AUROC indicates that the model can reliably separate normal from anomalous steps across different thresholds, and the high PR-AUC shows it remains precise even though anomalies are much less frequent. Taken together, these results suggest that the model is not only accurate but also sensitive to subtle deviations in daily routines.

Error analysis showed that most false positives occurred in Hygiene, Laying, Meals, and Sleep activities. These results suggest that activities with greater natural variability or contextual ambiguity are harder to classify consistently. For example, hygiene routines can vary considerably in timing and duration across individuals, while sleep and rest may include short naps or irregular awakenings that resemble anomalies. In contrast, mobility-related activities such as Walking or Standing produced very few errors, likely due to their distinct and temporally consistent patterns.

4 CONCLUSION AND FUTURE WORK

This work presents a reproducible approach for generating synthetic routines and injecting clinically relevant anomalies, enabling robust training of anomaly detection models. A GRU-based model achieved high accuracy, demonstrating the value of synthetic data in addressing data scarcity and ethical limitations. Future work will focus on validating with real-world wearable data, accounting for variability, and refining anomalies with clinical input.

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REFERENCES

- Alzheimer’s Association. (2025). *American Perspectives on Early Detection of Alzheimer’s Disease in the Era of Treatment*.
- Arantes, M., Carvalho, M., Rocha, I. C., Soares, J., Moreira, A., Pinto, M., Matos, D., Morais, P., & Carvalho, V. (2025). Towards the Development of a Wearable Device Enhancing Inclusivity and Monitoring for Dementia Patients. *Lecture Notes in Mechanical Engineering*, 360–370. https://doi.org/10.1007/978-3-031-94223-5_32
- Carvalho, M., Rocha, I., Arantes, M., Linhares, R., Soares, J., Moreira, A., Vilaça, J., Matos, D., Morais, P., & Carvalho, V. (2025). *Powered Wearable Technologies for Dementia Care: Evaluating Activity Recognition Models and Dataset Challenges*. 995–1006. <https://doi.org/10.5220/0013396600003911>
- Chalapathy, R., & Chawla, S. (2019). *Deep Learning for Anomaly Detection: A Survey*. <http://arxiv.org/abs/1901.03407>
- Freitas, A. R., Soares, J., Arantes, M., Rocha, I., Carvalho, M., Pinto, M., Matos, D., Morais, P., & Carvalho, V. (2025). *EAI Endorsed Transactions on Digital Transformation of Industrial Processes Wearable Devices in Dementia Monitoring: A Systematic Review of Technologies, Design, and Future Directions*. <https://doi.org/10.4108/1>
- Liu, X., Aksu, T., Liu, J., Wen, Q., Liang, Y., Xiong, C., Savarese, S., Sahoo, D., Li, J., & Liu, C. (2025). Empowering Time Series Analysis with Synthetic Data: A Survey and Outlook in the Era of Foundation Models. *Proceedings of Conference’17*, 1–11. <https://doi.org/10.1145/>
- Rocha, I. C., Arantes, M., Moreira, A., Vilaça, J. L., Morais, P., Matos, D., & Carvalho, V. (2024). Monitoring Wearable Devices for Elderly People with Dementia: A Review. In *Designs* (Vol. 8, Issue 4). Multidisciplinary Digital Publishing Institute (MDPI). <https://doi.org/10.3390/designs8040075>
- Smuck, M., Odonkor, C. A., Wilt, J. K., Schmidt, N., & Swiernik, M. A. (2021). The emerging clinical role of wearables: factors for successful implementation in healthcare. In *npj Digital Medicine* (Vol. 4, Issue 1). Nature Research. <https://doi.org/10.1038/s41746-021-00418-3>